Low-Rank Structure Is Geometry

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Short answer key · generated from the shared Julia exercise data

Every answer below is stored beside its question in `notebooks/ExerciseContent.jl`.

The goal is to distinguish the represented object, its coordinates, the model assumption, and the evidence needed before interpreting components.

Exercise 1 — Predict a multilinear map

notebooks/00_Primer.jl · Section 1 · about 6 minutes

Let T have size (4, 3, 3), with M1 of size (2, 4) and M2, M3 of size (2, 3). Predict before running the multilinear mode-product experiment.

1. Predict the size of Y = T ×1 M1 ×2 M2 ×3 M3.

□ (2, 2, 2) □ (4, 2, 2) □ (2, 3, 3) □ (4, 3, 3)

Answer: (2, 2, 2).

2. Predict the intermediate sizes after the mode-1 map and then the mode-2 map.

Answer: After mode 1: (2, 3, 3). After mode 2: (2, 2, 3).

3. Run the experiment. What final size does the notebook report?

Answer: (2, 2, 2).

4. Which original dimension is transformed independently by each of M1, M2, and M3?

Answer: M1 transforms the first dimension, M2 the second, and M3 the third.

5. How many entries does Y contain?

□ 6 □ 8 □ 12 □ 18

Answer: 8 entries.

6. What does this experiment show about a tensor as a multilinear object?

Answer: A separate linear map can act on each mode, replacing only that mode's dimension while the other mode structure remains explicit.

Exercise 2 — Build, decompose, reconstruct

notebooks/00_Primer.jl · Section 3 · about 7 minutes

Generate a rank-3 tensor from A of size (3, 3), B of size (3, 3), and C of size (2, 3). Fit a rank-3 CPD and inspect the returned object.

1. What is the size of the generated tensor, and how many rank-one terms were used?

Answer: The tensor has size (3, 3, 2) and uses three rank-one terms.

2. What is the format of the returned CP decomposition?

Answer: Three component weights and one factor matrix per mode; here the factor sizes are (3, 3), (3, 3), and (2, 3).

3. Reconstruct T-hat. What size does it have?

Answer: (3, 3, 2).

4. Record ||T - T-hat||F / ||T||F from the notebook.

Answer: Accept the near-zero relative Frobenius error printed by the notebook.

5. If the relative error is nearly zero, what has been verified?

Answer: The represented tensor has been reconstructed accurately.

6. What has not been verified by a near-zero error?

Answer: Uniqueness, stability, and recovery of the exact original factor coordinates have not been verified.

7. Must the fitted factor columns equal the original columns entry by entry?

□ Yes □ No

Answer: No. CP coordinates can differ by component permutation and reciprocal rescaling even when the tensor is identical.

Exercise 3 — Same object, different coordinates

notebooks/01_OneObjectManyCoordinates.jl · Sections 1A–1B · about 8 minutes

Open the Matrix gauge experiment. Make a prediction, reveal the experiment, and then move the gauge slider.

Measure

1. Set log10(s) = 0. Record s and κ(Q).

Answer: s = 1 and κ(Q) = 1.

2. Set log10(s) = 3. Record s and κ(Q).

Answer: s = 1000 and κ(Q) = 1,000,000.

3. Tick every quantity that changes strongly.

□ factor coordinates □ factor conditioning □ represented matrix

Answer: Factor coordinates and factor conditioning change strongly; the represented matrix does not.

4. In the CP rescaling and permutation experiment, does the represented tensor change?

□ Yes □ No

Answer: No.

Predict and explain

5. For $Q(s) = \mathrm{diag}(s, s^{-1})$, what do you expect as s becomes very large?

□ X changes dramatically □ the factors may become badly scaled □ the reconstruction error stays near zero □ κ(Q) increases

Answer: The factors can become badly scaled, the reconstruction error stays near zero, and κ(Q) increases.

6. If $A' = AQ$ and $B' = BQ^{-T}$, complete $A'B'^{T}$ = ____.

Answer: $A'B'^{T} = AB^{T} = X$.

7. Two different factor pairs produce the same matrix. Which object should we regard as the underlying object?

□ A □ B □ the pair (A, B) □ the represented matrix X

Answer: The represented matrix X.

8. If $X = A_1B_1^{T} = A_2B_2^{T}$, must $A_1 = A_2$ and $B_1 = B_2$?

□ Yes □ No

Answer: No.

9. A relative reconstruction error is 10^{-14}. Which conclusion is justified?

□ the factors are unique □ the recovered matrix is extremely close □ the problem is well-conditioned □ every factor was recovered correctly

Answer: The recovered matrix is extremely close to the target matrix; this alone says nothing about factor uniqueness or conditioning.

10. Finish the sentence: A low-rank factorization is a ____ of the underlying matrix. The coordinates can change while the represented ____ stays the same.

Answer: A representation (or coordinate description) of the underlying matrix; the represented object stays the same.

Exercise 4 — Multilinear rank and model choice

notebooks/02_GeometryAtlas.jl · HOSVD check and model cards · about 8 minutes

Compute the HOSVD summary of the supplied random tensor of size (3, 3, 2), then compare CP, Tucker, and BTD on the two-block target.

Multilinear rank

1. Record the sizes of the three unfoldings.

Answer: (3, 6), (3, 6), and (2, 9).

2. Record their matrix ranks. What is the tensor's multilinear rank?

Answer: The ranks are (3, 3, 2), so the multilinear rank is (3, 3, 2).

3. Why can the third multilinear rank never exceed 2 for a tensor of size (3, 3, 2)?

Answer: Its mode-3 unfolding has only two rows, so its matrix rank is at most 2.

4. Can a tensor have CP rank 3 while its third multilinear rank is only 2?

□ Yes □ No

Answer: Yes. CP rank and multilinear rank measure different structures.

Choose the structural model

5. Which model uses one shared list of rank-one components across all modes?

□ CP □ Tucker □ BTD

Answer: CP.

6. Which model uses one core with a possibly different compression rank for each mode?

□ CP □ Tucker □ BTD

Answer: Tucker.

7. The Lab 2 target is a sum of two small Tucker blocks. Which model most directly mirrors that construction?

□ CP □ Tucker □ BTD

Answer: BTD.

8. If one fitted model has the smallest error in this finite run, does that prove it is always the best model?

□ Yes □ No

Answer: No. Structural capacity, optimization success, numerical error, and the scientific meaning of the coordinates are different issues.

Exercise 5 — Observe stagnation, diagnose its geometry

notebooks/03_OptimizationFailureMuseum.jl · about 10 minutes

Use the failure map, collision-distance dial, three-case comparison, solver race, and plateau microscope to separate an observation from its possible cause.

Collision and the ALS Gram matrix

1. Set $\rho = 0.90$. What is the sign-invariant rank-one collision distance?

Answer: The overlap is $\rho^3 = 0.729$, so $d = \sqrt{(2 - 2\rho^3)} \approx 0.736$.

2. At $\rho = 0.90$, what are the off-diagonal entries of the two factor Gram matrices and of the ALS Gram matrix?

Answer: The factor Gram entries are $\rho = 0.90$; the ALS Gram entry is $\rho^2 = 0.81$.

3. As ρ approaches 1, which eigenvalue becomes small: $1 + \rho^2$ or $1 - \rho^2$?

□ $1 + \rho^2$ □ $1 - \rho^2$

Answer: $1 - \rho^2$.

4. In plain language, what becomes hard for ALS when the [1, −1] direction nearly disappears?

Answer: ALS can still identify the combined contribution, but it has difficulty deciding how much belongs to each of the two nearly identical components.

5. Does this collision example require large opposite weights?

□ Yes □ No

Answer: No. The main experiment uses ordinary weights (1, 1, 0.7); cancellation is a separate bonus effect.

Trajectory and plateau diagnosis

6. Why must every solver in the race start from the same CPDPoint?

Answer: It isolates the optimization method as the changed variable; otherwise different initial coordinates could explain different outcomes.

7. What observation triggers the plateau flag, and which diagnostics are used afterward to test a collision explanation?

Answer: The flag uses only log improvement below 0.05 over the last 20 sweeps. Minimum rank-one distance and ALS Gram conditioning are diagnostic evidence, not part of the plateau detector.

8. The poor-start case is slow even though its component distance is large and its ALS condition number is moderate. Does slow optimization by itself prove collision?

□ Yes □ No

Answer: No. A plateau is an observation; initialization and other mechanisms can explain it.

9. Can a reconstruction error be small while individual components remain difficult to interpret reliably?

□ Yes □ No

Answer: Yes. Object-level fit and coordinate-level separation or stability are different diagnostics.

Exercise 6 — From activations to a concept claim

notebooks/04_NeuralRepresentations.jl · Concept explorer and evidence ladder · about 14 minutes

Audit one candidate concept in the synthetic CRAFT-style pipeline. Separate what the factorization shows from the semantic and behavioral claims that still require evidence.

Read the factorization

1. What does one row of the activation matrix A represent?

Answer: The pooled activation-feature vector produced by one image crop.

2. In $A \approx UW^T$, what do U[i,j] and W[:,j] represent?

Answer: U[i,j] is candidate-concept j's nonnegative usage in crop i; W[:,j] is its candidate concept activation direction in feature space.

3. Move the rank slider from 1 to 5. At which rank are the three planted ingredients easiest to name, and what happens on either side?

Answer: They are clearest near rank 3. Smaller ranks merge ingredients; larger ranks can split or duplicate them even while reconstruction error falls.

4. Choose one concept. Which evidence helps you propose a human label: the CAV alone or the high-usage crops?

□ CAV alone □ High-usage crops

Answer: The high-usage crops connect the activation direction to recurring visible examples; the CAV alone is not human-readable.

Audit the claim

5. In the perturbation experiment, is the most present concept necessarily the most important to the class score?

□ Yes □ No

Answer: No. Presence is a coordinate in the activation reconstruction; importance concerns the model output's sensitivity to changing that coordinate.

6. Why is nonnegativity not enough for a concept claim? Give one stability check.

Answer: NMF coordinates can depend on rank, layer, initialization, and optimization. For example, compare factors and top crops across seeds, nearby ranks, or layers.

7. Why is the sample × location × feature factorization in 4H not CRAFT?

Answer: CRAFT factorizes pooled crop activations with matrix NMF and uses its own attribution machinery for location. The tensor model is a TensorKitchen extension question with different structural assumptions.